

Project Summary

Clinical deterioration in hospitalized children remains a significant patient safety challenge and is associated with increased morbidity, unplanned pediatric intensive care unit (PICU) transfers, prolonged hospitalization, and mortality. Although Pediatric Early Warning Systems (PEWS) are widely used to support recognition of deterioration and guide escalation of care, current approaches remain primarily reactive and provide limited ability to identify children at risk before substantial physiologic decline occurs. Consequently, opportunities for early intervention and prevention of adverse events may be missed.

The long-term goal of this research is to improve pediatric patient safety through earlier recognition and prevention of clinical deterioration. The objective of this project is to develop and validate a predictive pediatric early warning framework that integrates longitudinal PEWS trajectories and routinely collected clinical data to forecast clinical deterioration before conventional escalation thresholds are reached. The central hypothesis is that temporal patterns in PEWS scores, vital signs, and other clinical variables contain early physiologic signatures that can be leveraged to generate clinically actionable deterioration forecasts.

To achieve this objective, we will pursue three aims: (1) develop and validate predictive models for pediatric deterioration forecasting using longitudinal PEWS trajectories and clinical variables; (2) identify clinically actionable forecasting horizons and alert thresholds through multi-horizon evaluation of predictive performance, lead time, and alert burden; and (3) evaluate the interpretability, clinical plausibility, and potential actionability of predictive deterioration forecasts through explainable artificial intelligence methods and structured review by pediatric clinical experts. The project will utilize retrospective pediatric inpatient data, longitudinal trajectory modeling, machine learning methods, and clinician-centered evaluation of predictive forecasts.

Expected outcomes include validated predictive models, evidence-based forecasting horizons and alert thresholds, and a clinician-informed predictive deterioration framework suitable for future prospective validation and implementation. Successful completion of this project will establish the scientific and translational foundation for predictive pediatric early warning systems capable of enabling earlier intervention, reducing preventable adverse events, decreasing unplanned PICU transfers, and improving outcomes for hospitalized children.

Specific Aims

Clinical deterioration in hospitalized children remains a major patient safety challenge and a leading cause of unplanned intensive care unit (ICU) transfer, cardiopulmonary arrest, prolonged hospitalization, and preventable morbidity. Pediatric Early Warning Systems (PEWS) are widely used to support recognition of deterioration and guide escalation of care; however, current PEWS frameworks remain fundamentally reactive, relying on threshold-based assessment of a child's current physiologic state rather than forecasting future risk. Emerging evidence and our preliminary studies demonstrate that measurable physiologic changes occur hours before clinical decompensation. Nevertheless, existing PEWS frameworks do not leverage these longitudinal trajectories to anticipate future deterioration. *Thus, a critical unmet need exists for clinically deployable methods capable of identifying vulnerable pediatric patients before traditional PEWS escalation thresholds are reached, thereby providing clinicians with sufficient time to intervene and prevent adverse outcomes.*

The long-term goal of this research is to improve pediatric patient safety through earlier recognition and prevention of clinical deterioration. The objective of this proposal is to develop and validate a predictive deterioration forecasting framework that integrates longitudinal PEWS trajectories and routinely collected clinical data to identify children at risk before overt physiologic decline occurs. Our central hypothesis is that temporal patterns in PEWS scores, vital signs, and clinical characteristics encode early physiologic signatures of deterioration that can be detected several hours before conventional PEWS thresholds are reached and translated into clinically actionable risk forecasts. To test this hypothesis, we will develop and evaluate machine learning and deep learning models that transform routinely collected clinical data into predictive deterioration forecasts. This hypothesis is supported by our preliminary work conducted through an ongoing collaboration with Children's Health Dallas. Using an existing dataset of 15,276 de-identified pediatric inpatient encounters, including 184 deterioration events requiring ICU transfer, we have established a predictive modeling pipeline and demonstrated the feasibility of AI-based deterioration forecasting. Children's Health Dallas has committed to providing an expanded pediatric inpatient dataset to support model refinement and validation. Combined with our multidisciplinary expertise in pediatric nursing, pediatric clinical care, and artificial intelligence, these resources uniquely position our team to accomplish the proposed work. To achieve our objective, we propose the following Specific Aims:

Aim 1. Develop and validate predictive models for pediatric deterioration forecasting. We will integrate longitudinal PEWS trajectories, vital signs, demographic characteristics, and routinely collected clinical variables to develop predictive models capable of forecasting pediatric clinical deterioration. We hypothesize that temporal patient trajectories contain predictive information that significantly improves deterioration forecasting compared with conventional threshold-based PEWS monitoring. The expected outcome is a validated predictive modeling framework that accurately forecasts pediatric deterioration risk before ICU transfer and supports subsequent evaluation of clinically actionable forecasting strategies.

Aim 2. Identify clinically actionable forecasting horizons and alert thresholds. Using the predictive models developed in Aim 1, we will evaluate predictive performance across 3-, 6-, 9-, 12-, 15- and 18-hour forecasting horizons and determine risk thresholds that optimize the balance between predictive accuracy, intervention lead time, calibration, and alert burden. We hypothesize that clinically meaningful deterioration forecasts can be generated sufficiently early to support proactive intervention while maintaining acceptable predictive performance and manageable alert frequency. The expected outcome is identification of evidence-based forecasting horizons and alert thresholds that maximize clinical utility.

Aim 3. Establish an interpretable and clinician-informed predictive deterioration framework. Building upon the predictive models developed in Aim 1 and the candidate alert rules identified in Aim 2, we will evaluate the interpretability, clinical plausibility, and potential actionability of deterioration forecasts through explainable artificial intelligence methods and structured review of de-identified patient cases by pediatric clinical experts. We hypothesize that predictive deterioration forecasts accompanied by patient-specific explanation reports will be judged as clinically plausible, interpretable, and potentially actionable within existing pediatric inpatient workflows. The expected outcome is a clinician-informed predictive deterioration framework that combines predictive accuracy, actionable forecasting horizons, and interpretable risk communication strategies suitable for future prospective validation and implementation.

The proposed research is innovative because it transforms PEWS from a reactive deterioration detection tool into a predictive deterioration forecasting framework. By integrating longitudinal PEWS trajectories, multi-horizon forecasting, and interpretable artificial intelligence, this project moves beyond traditional threshold-based monitoring and current-state risk assessment to provide actionable forecasts of future deterioration risk. Successful completion of this project will yield validated predictive models, evidence-based forecasting horizons and alert thresholds, and a clinician-informed decision-support framework for pediatric deterioration surveillance. These outcomes will establish the scientific and translational foundation for a new generation of predictive pediatric early warning systems capable of enabling earlier intervention, improving patient safety, and transforming deterioration surveillance from reactive detection to proactive prevention.